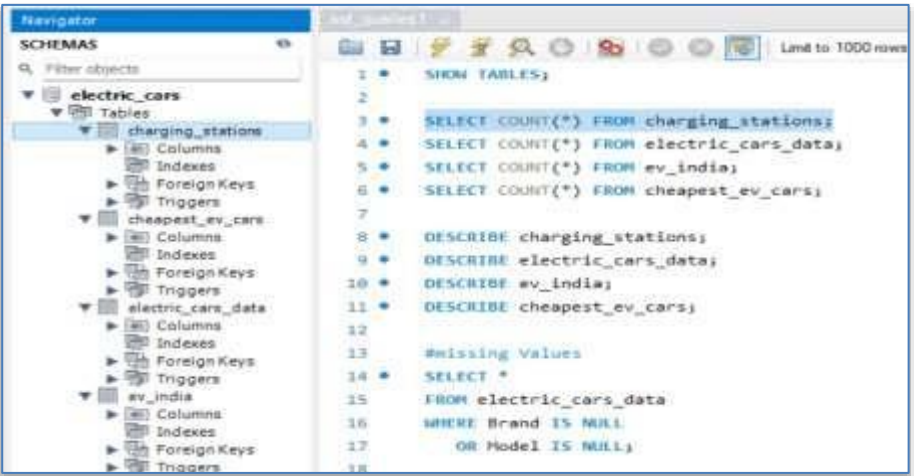
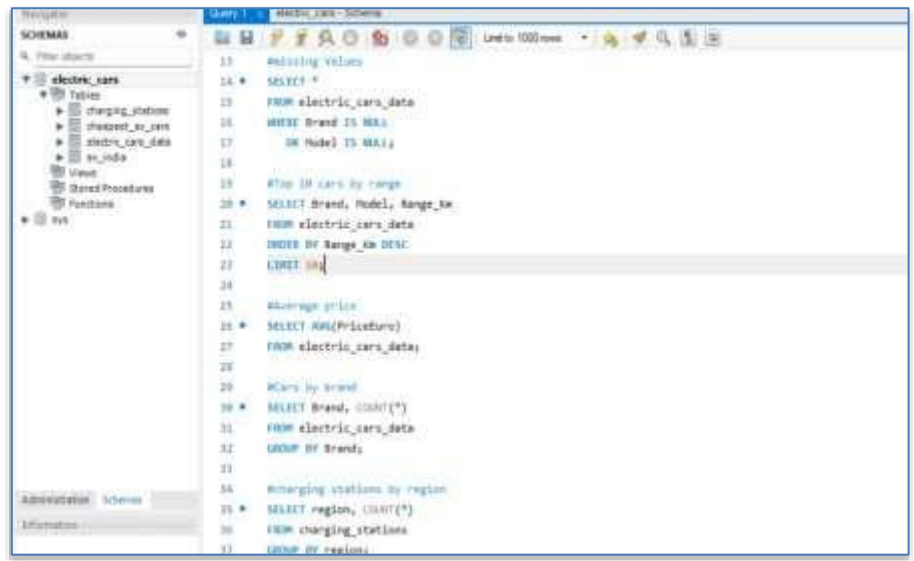
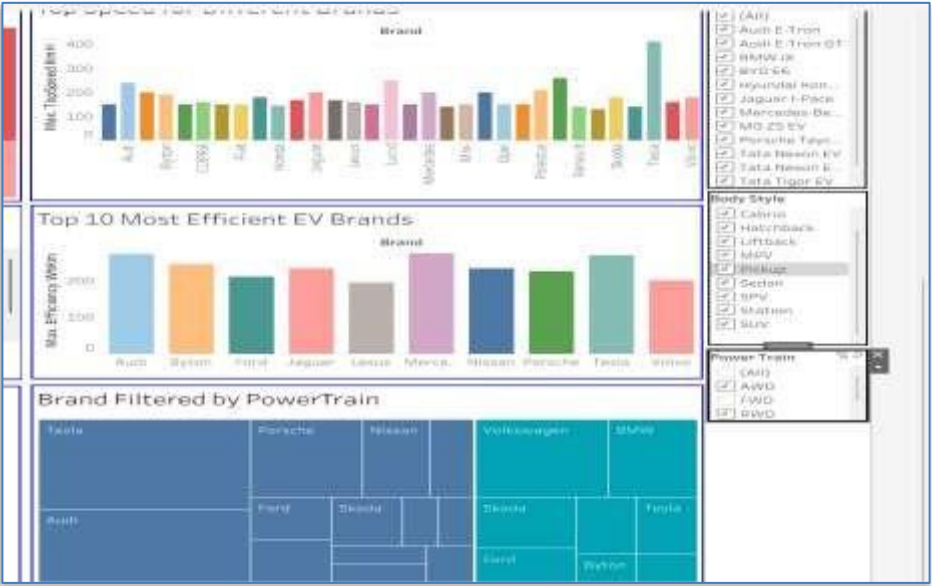
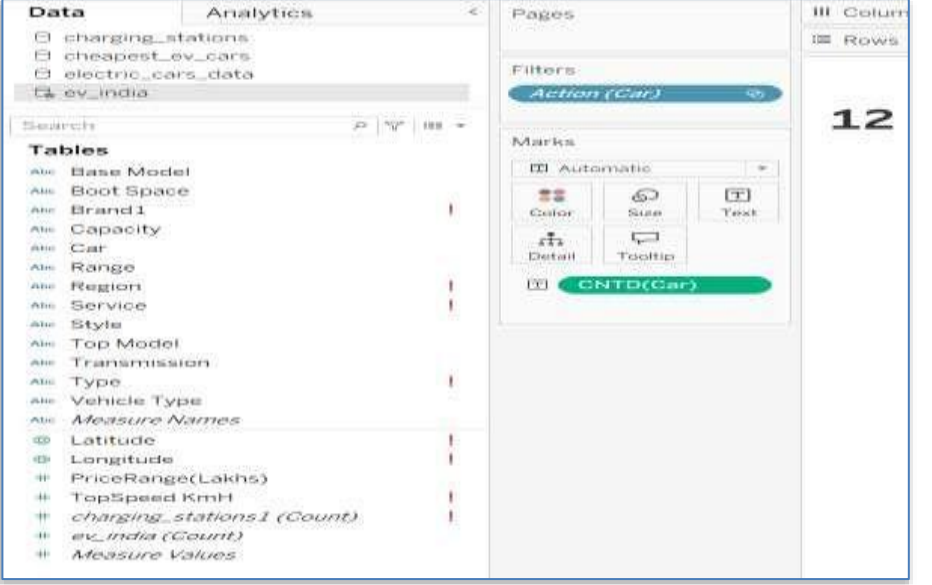


Project Performance Test

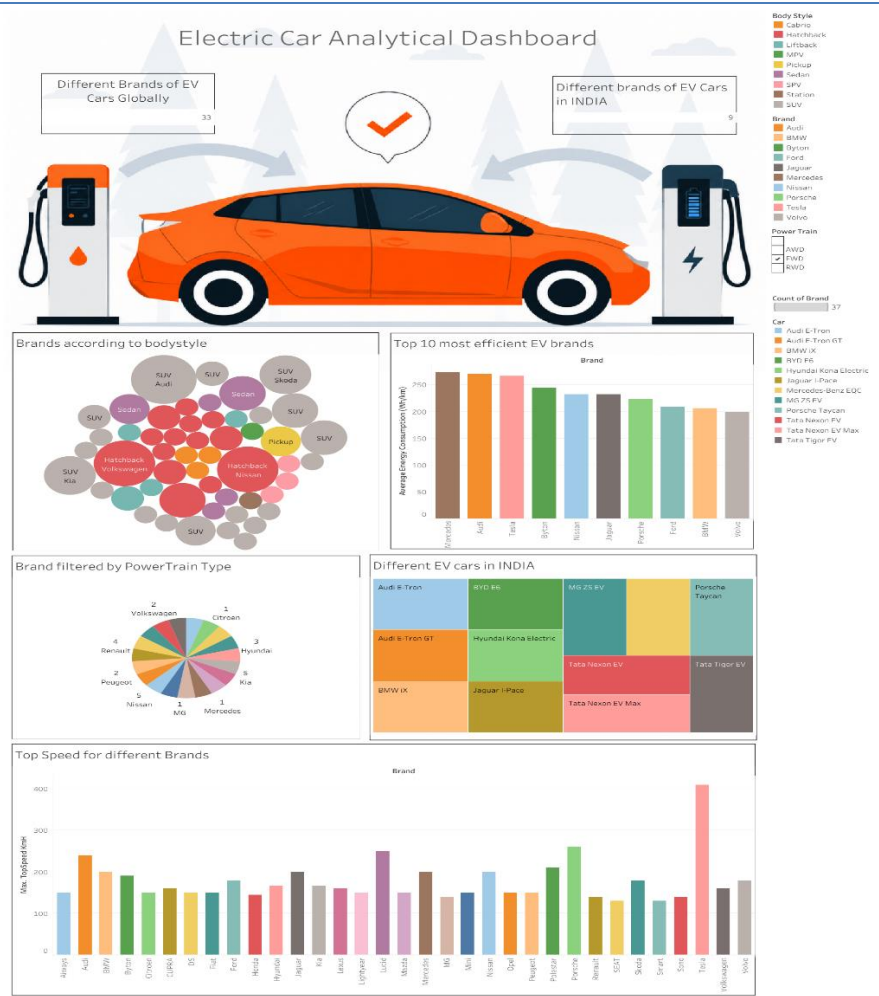
Date	08 July 2026
Team ID	SWTID-2026-4616
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Total 4 EV datasets successfully rendered from MySQL database into Tableau Desktop.  <pre> 1 * SHOW TABLES; 2 3 * SELECT COUNT(*) FROM charging_stations; 4 * SELECT COUNT(*) FROM electric_cars_data; 5 * SELECT COUNT(*) FROM ev_india; 6 * SELECT COUNT(*) FROM cheapest_ev_cars; 7 8 * DESCRIBE charging_stations; 9 * DESCRIBE electric_cars_data; 10 * DESCRIBE ev_india; 11 * DESCRIBE cheapest_ev_cars; 12 13 #missing values 14 * SELECT * 15 FROM electric_cars_data 16 WHERE Brand IS NULL 17 OR Model IS NULL; 18 </pre>
2.	Data Preprocessing	Data preprocessing completed before visualization: • Removed duplicate records • Checked missing values • Standardized column names • Converted numerical fields • Imported cleaned datasets into MySQL  <pre> 13 #missing values 14 * SELECT * 15 FROM electric_cars_data 16 WHERE Brand IS NULL 17 OR Model IS NULL; 18 19 #Top 10 cars by range 20 * SELECT Brand, Model, Range_Km 21 FROM electric_cars_data 22 ORDER BY Range_Km DESC 23 LIMIT 10; 24 25 #Average price 26 * SELECT AVG(PriceEuro) 27 FROM electric_cars_data; 28 29 #Cars by brand 30 * SELECT Brand, COUNT(*) 31 FROM electric_cars_data 32 GROUP BY Brand; 33 34 #Charging stations by region 35 * SELECT region, COUNT(*) 36 FROM charging_stations 37 GROUP BY region; </pre>

3.	Utilization of Filters	Interactive Tableau filters implemented:  The image displays three Tableau bar charts illustrating the utilization of filters. The top chart, titled 'Brand', shows 'Max. Top Speed (km/h)' on the y-axis (0 to 400) and 'Brand' on the x-axis. The middle chart, titled 'Top 10 Most Efficient EV Brands', shows 'Max. Efficiency (Wh/km)' on the y-axis (0 to 2000) and 'Brand' on the x-axis. The bottom chart, titled 'Brand Filtered by PowerTrain', is a treemap showing brands categorized by 'Power Train'. To the right of the charts are two filter shelves. The 'Body Style' shelf includes options like Cabrio, Hatchback, Liftback, Sedán, Sedán, SUV, and SUV. The 'Power Train' shelf includes options like AWD, AWD, and AWD. The bottom chart is filtered to show only AWD vehicles.
4.	Calculation fields Used	Calculated fields created for analytical insights:  The image shows the Tableau interface with the 'Data' pane on the left, the 'Analytics' pane in the center, and the 'Columns' and 'Rows' shelves on the right. The 'Data' pane lists several tables: 'charging_stations', 'cheapest_lev_cars', 'electric_cars_data', and 'ev_india'. The 'Analytics' pane shows a search bar and a list of fields: 'Base Model', 'Boot Space', 'Brand1', 'Capacity', 'Car', 'Range', 'Region', 'Service', 'Style', 'Top Model', 'Transmission', 'Type', 'Vehicle Type', 'Measure Names', 'Latitude', 'Longitude', 'PriceRange(Lakhs)', 'TopSpeed KmH', 'charging_stations1 (Count)', 'ev_india (Count)', and 'Measure Values'. The 'Columns' shelf contains the calculated field 'CNTD(Car)' and the 'Rows' shelf contains the calculated field '12'.

Dashboard design



Story Design

